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Supplementary Notes

Hexagulation numbers: symmetry breaking on the gyroid surface

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I. HEXAGULATION NUMBER

Caspar and Klug showed that icosahedral shells can be constructed from twenty triangles, with the triangulation number T taking integer values such as 1, 3, 4, 7, 9, 12, ..., given by

$$T = h^2 + k^2 + hk \tag{1}$$

where h and k are nonnegative integers. Figure 1a shows the T-diagram. A lattice point labeled (h, k) (with $\mathbf{L} = h\mathbf{e}_1 + k\mathbf{e}_2$) in the oblique coordinate system spanned by basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ on the triangular lattice indicates the circled number T, defined as half the number of lattice points (or equivalently the number of triangles) inside the triangle whose side is the segment between the origin and the point (h, k). Considering the large triangle indicated by the solid line, with lattice points located at its vertices, the number of vertices N of the entire icosahedral shell is given by $N = 10T + 2$.

By analogy, we define the "hexagulation number" H as the number of dashed hexagons inside a large hexagon (solid line) in the H-diagram shown in Fig. 1b. Let $\mathbf{L} = h\mathbf{e}_1 + k\mathbf{e}_2$ be a vector along an edge of the large hexagon, where $\mathbf{e}_1$ and $\mathbf{e}_2$ are basis vectors of the oblique coordinate system. We then obtain $H = |\mathbf{L}|^2$, giving

$$H = h^2 + k^2 - hk, \tag{2}$$

which produces the sequence $1, 3, 4, 7, 9, 12, \ldots$ for integer pairs (h, k). Note that the only difference between T and H is the sign of the third term.

According to space-group symmetries, the triply periodic minimal surfaces can be constructed from 8, 16, or 32 hexagonal patches per conventional cubic unit cell for the primitive (P), gyroid (G), and diamond (D) surfaces, respectively. Therefore, the number of spheres per unit cell is $N_P = 8H$, $N_G = 16H$, $N_D = 32H$ for the P-, G-, and D-surfaces, respectively. In this paper, we focus on the gyroid surface and perform simulations for higher values of H up to $H = 21$.

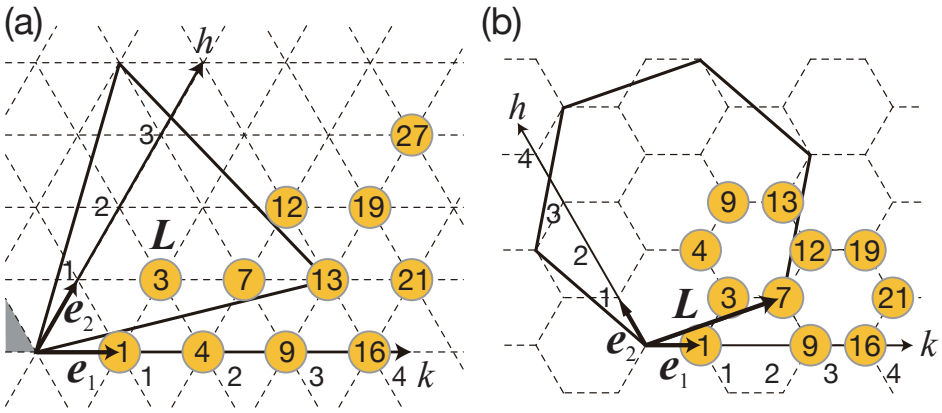


Figure 1. (**a**) Caspar and Klug's T-diagram for counting the triangulation number T, and (**b**) the H-diagram for counting the hexagulation number H. Lattice points labeled (h, k) with nonnegative integers h and k in two oblique coordinate systems indicate the circled numbers $T = h^2 + k^2 + hk$ and $H = h^2 + k^2 - hk$: T is half the number of lattice points inside a triangle (solid line), whereas H is the number of dashed hexagons inside a hexagon (solid line). Shown here are $T = 13$ for $(3, 1)$ and $H = 7$ for $(3, 1)$.

II. SUPPLEMENTARY TABLES

TABLE I. Regular tessellations of hard spheres on the G-surface, classified by the hexagulation number H, the total number of spheres in a unit cell $N = 16H$. Space groups of the ordered structures are presented. When H is a multiple of three, the centrosymmetric space group $Ia\bar{3}d$ is broken to the non-centrosymmetric space group $I\bar{4}3d$. Wyckoff positions are denoted as mw, representing multiplicity m and Wyckoff letter w corresponding to the space groups and polygonal tilings indicated by the vertex. The set of integers $(n_1.n_2.n_3.\cdots; n_4.n_5.n_6.\cdots;)$ denotes the local tiling around a vertex, where n_1-gon, n_2-gon, and n_3-gon, $\cdots$ meet consecutively. Superscripts are used for abbreviations when possible. Polygonal tilings are constructed from the sphere-center positions. Note that $H = 4$ class illustrates a hypothetical high-symmetry structure, which is not obtained in our simulation. See details in Ref. [1].

H-number	Surface (N)	Space group	Wyckoff position	Tiling
$H = 3$	G (48)	$I\bar{4}3d$ (No. 220)	48e	$(3^3.4.3.4)$
$H = 4$	G (64)	$Ia\bar{3}d$ (No. 230)	16a; 48g	$(3^6; 3^2.4.3^2.4)$
$H = 7$	G (112)	$Ia\bar{3}d$ (No. 230)	16a; 96h	$(3^6; 3^5.4)$
$H = 9$	G (144)	$I\bar{4}3d$ (No. 220)	48e×3	$(3^6; 3^5.4; 3^5.4)$
$H = 12$	G (192)	$I\bar{4}3d$ (No. 220)	48e×4	$(3^6; 3^6; 3^5.4; 3^5.4)$
$H = 13$	G (208)	$Ia\bar{3}d$ (No. 230)	16a; 96h; 96h;	$(3^6; 3^5.4; 3^2.4.3.4)$
$H = 16$	G (256)	$Ia\bar{3}d$ (No. 230)	16a; 48g; 96h; 96h;	$(3^6; 3^4.4; 3^2.4.3.4; 3^2.4.3.4)$
$H = 19$	G (304)	$Ia\bar{3}d$ (No. 230)	16a; 96h; 96h; 96h;	$(3^6; 3^6; 3^6; 3^5.4)$
$H = 21$	G (336)	$I\bar{4}3d$ (No. 220)	48e×7	$(3^6; 3^6; 3^6; 3^6; 3^6; 3^5, 4; 3^5.4)$

TABLE II. Explicit coordinates in terms of Wyckoff positions for the regular tessellations shown in Table 1. Note that special points having exact values given in the International Tables of Crystallography are omitted. In real simulations, $H = 4$ provides the lower symmetry configuration.

H-number	Surface (N_a)	Space group	Wyckoff p.	x	y	z
$H = 3$	G (48)	$I\bar{4}3d$ (No. 220)	48e	0.576	0.386	0.554
$H = 4$	G (64)	$Ia\bar{3}$ (No. 206)	48e	0.067	0.655	0.147
$H = 7$	G (112)	$Ia\bar{3}d$ (No. 230)	96h	0.039	0.144	0.411
$H = 9$	G (144)	$I\bar{4}3d$ (No. 220)	48e	0.813	0.240	0.300
			48e	0.010	0.650	0.604
			48e	0.064	0.339	0.355
$H = 12$	G (192)	$I\bar{4}3d$ (No. 220)	48e	0.7924	0.8535	0.5766
			48e	0.9068	0.7814	0.6245
			48e	0.2980	0.2753	0.1801
			48e	0.5857	0.8865	0.4549
$H = 13$	G (208)	$Ia\bar{3}d$ (No. 230)	96h	0.4436	0.3152	0.1064
			96h	0.6651	0.2274	0.1570
$H = 16$	G (256)	$Ia\bar{3}d$ (No. 230)	96h	0.2410	0.3652	0.9332
			96h	0.9255	0.5916	0.0282
$H = 19$	G (304)	$Ia\bar{3}d$ (No. 230)	96h	0.3679	0.0551	0.3888
			96h	0.7847	0.1459	0.9207
			96h	0.9891	0.4172	0.4349
$H = 21$	G (336)	$I\bar{4}3d$ (No. 220)	48e	0.4613	0.0038	0.0438
			48e	0.6138	0.1899	0.1090
			48e	0.1817	0.1130	0.5155
			48e	0.5827	0.5365	0.3685
			48e	0.9480	0.0862	0.3796
			48e	0.2854	0.6193	0.5628
			48e	0.7520	0.6739	0.1611

[1] Dotera T, Tanaka H, Takahashi Y (2017) Hexagulation numbers: the magic numbers of equal spheres on triply periodic minimal surfaces. Struct. Chem. 28:105-112